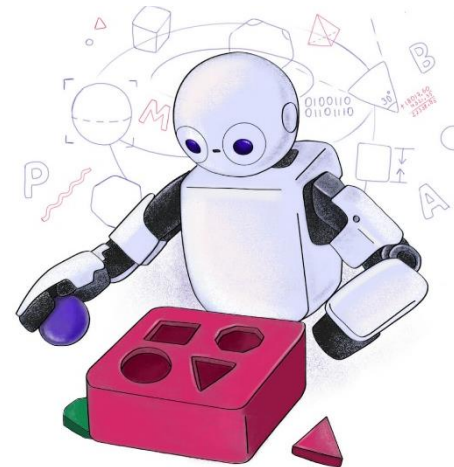


TP557 - Tópicos avançados em IoT e Machine Learning: *Arduino nano BLE 33 sense*

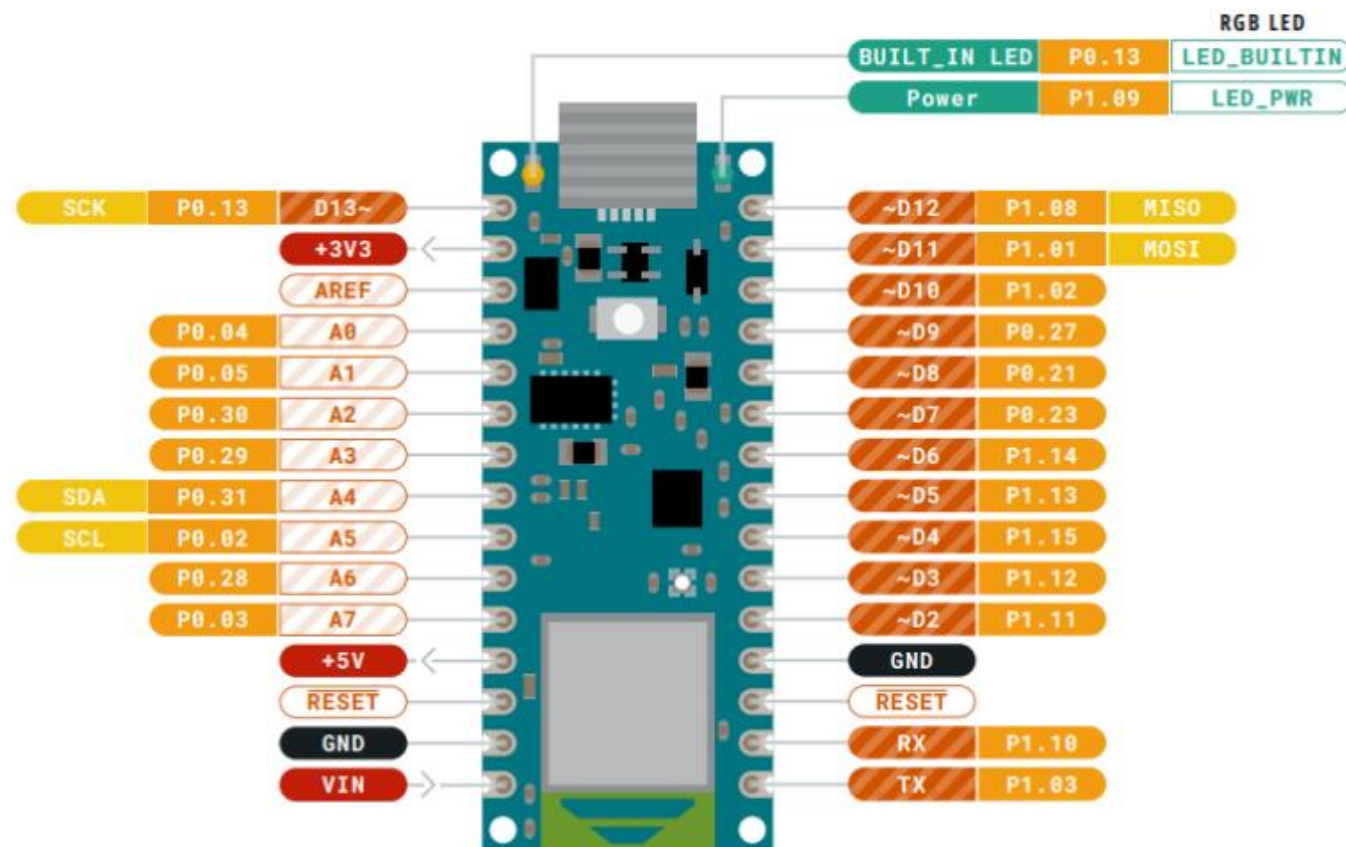
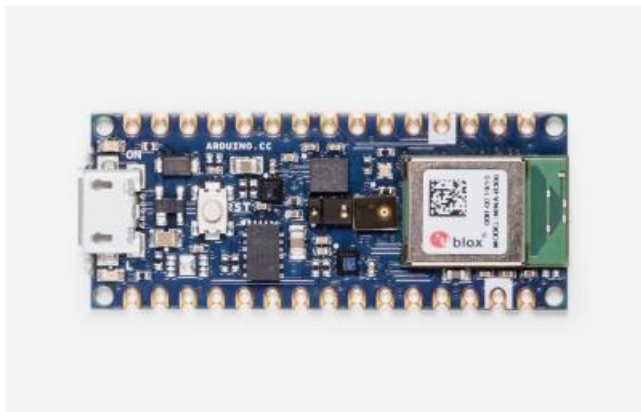


Inatel

Créditos: Prof. Marcelo Rovai UNIFEI

Samuel Baraldi Mafra
samuelbmafra@inatel.br

Arduino nano BLE 33 sense



Pinout

Arduino nano BLE 33 sense

Sensores e protocolo



Bluetooth®



IMU for Motion Detection

unidade de medição inercial (IMU):
acelerômetro, giroscópio e
magnetômetro



Microphone



**Proximity and Gesture
Detection**



**Barometric Pressure
Sensor**

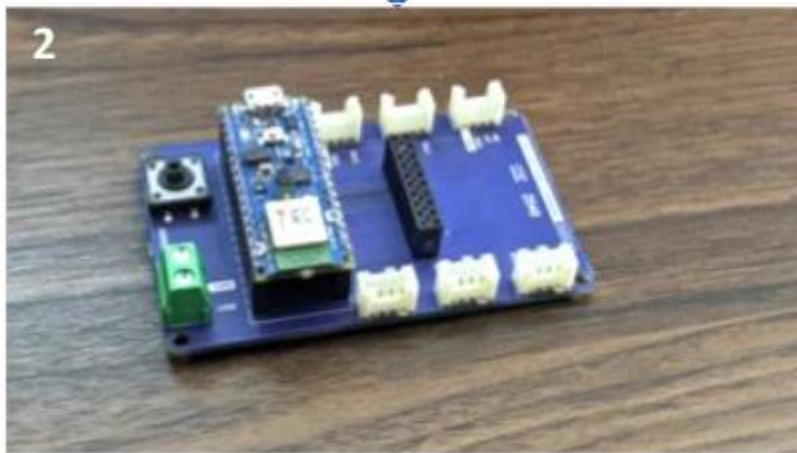
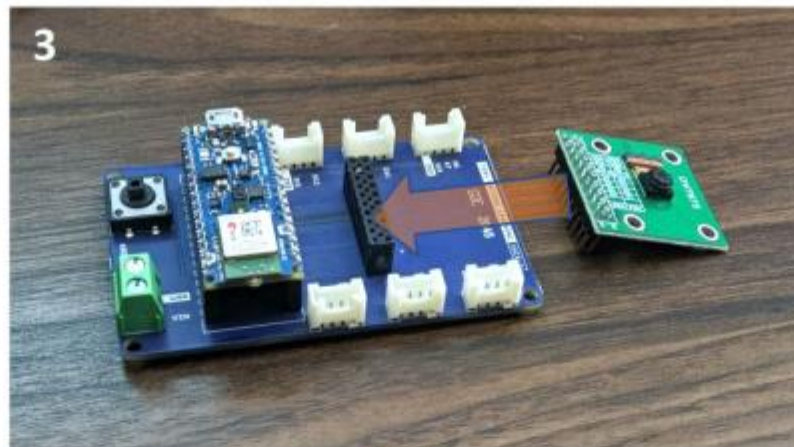
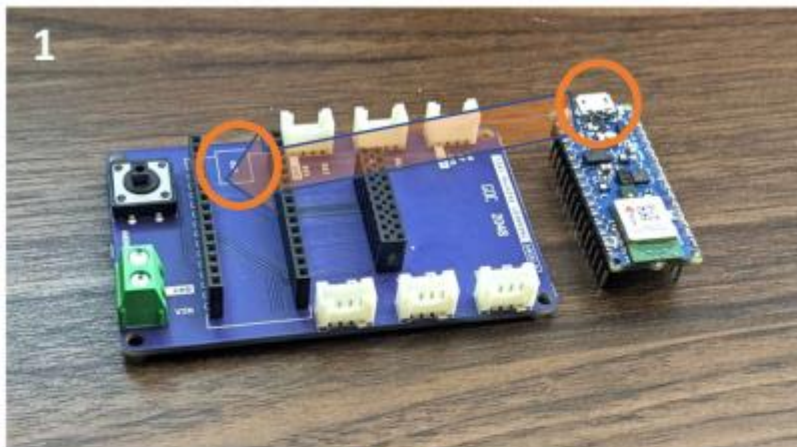


**Temperature and
*Humidity Sensor**

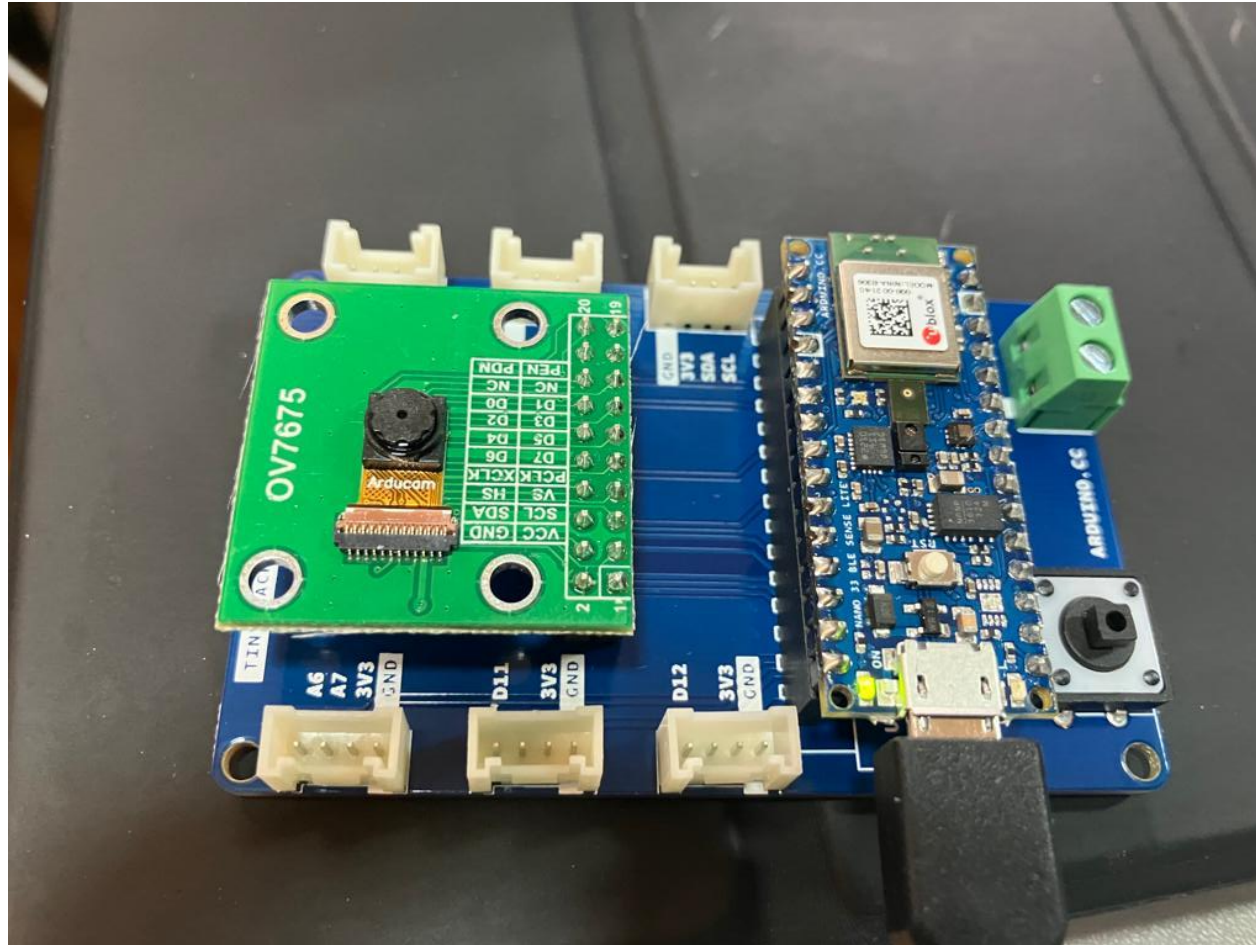
*Não está instalado em algumas versões.

Arduino nano BLE 33 sense

Instalando o hardware

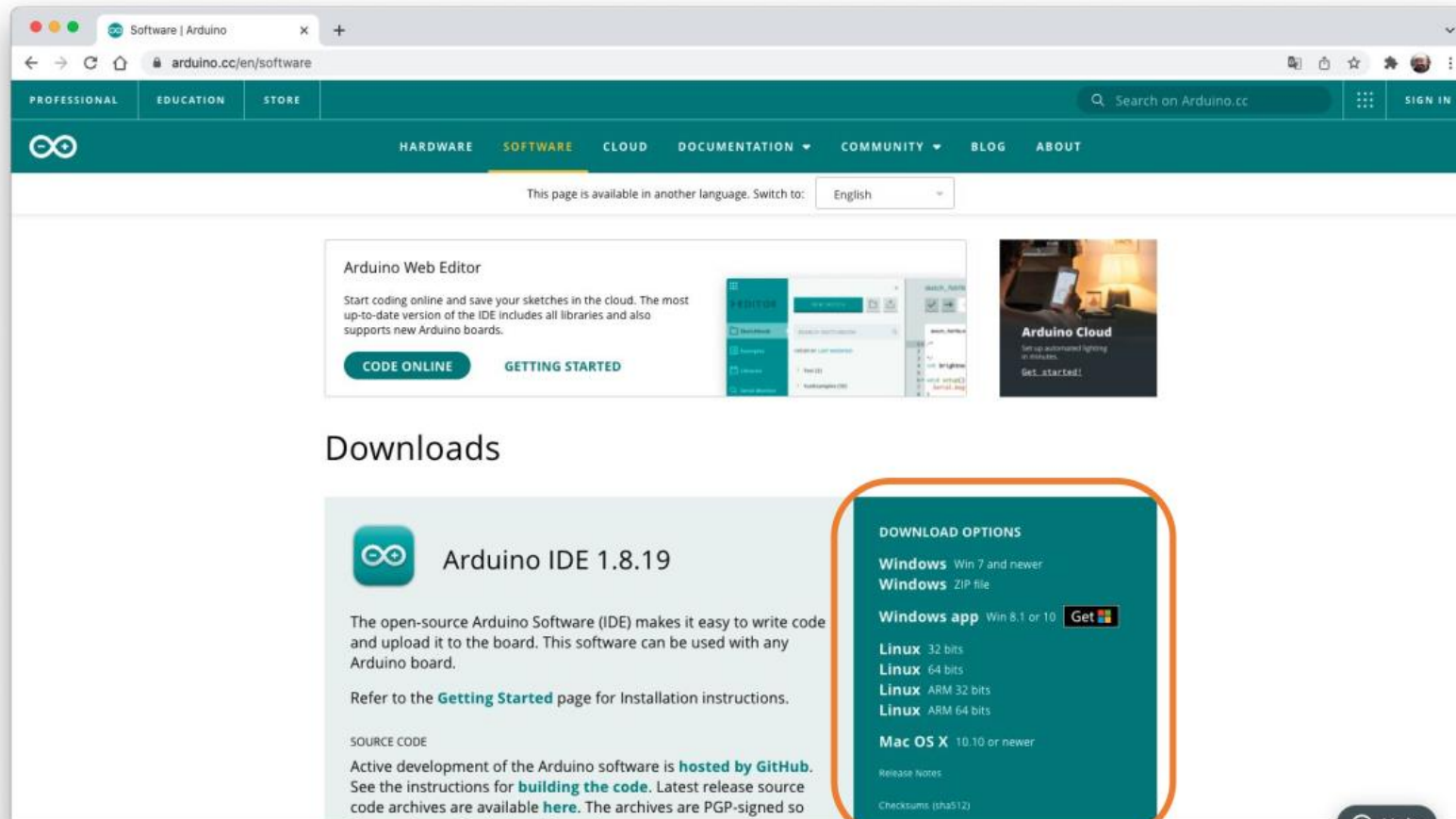


Arduino nano BLE 33 sense



Arduino nano BLE 33 sense

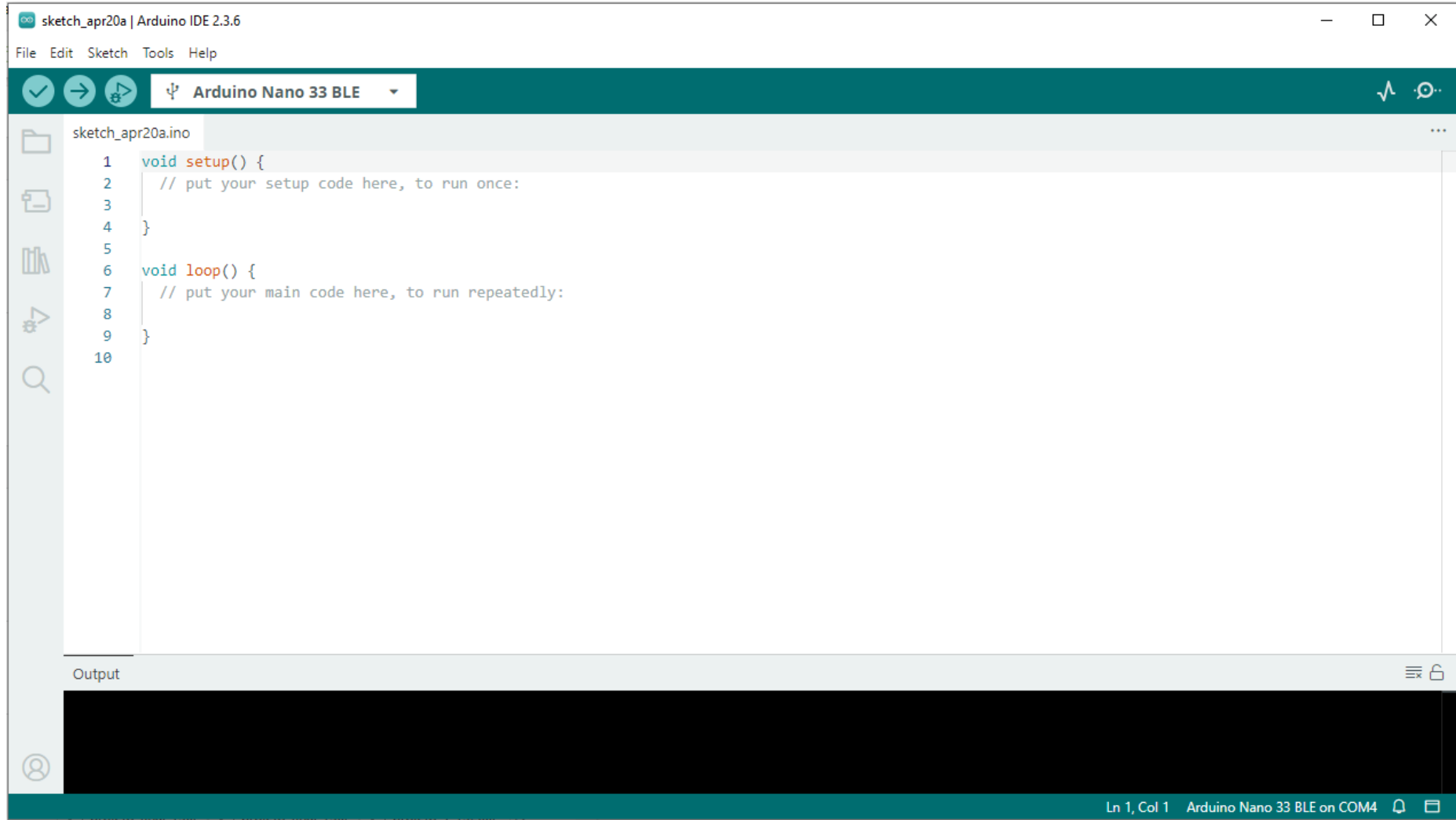
Installing the Arduino IDE



<https://www.arduino.cc/en/software/>

Arduino nano BLE 33 sense

Menus e
barra de
tarefas

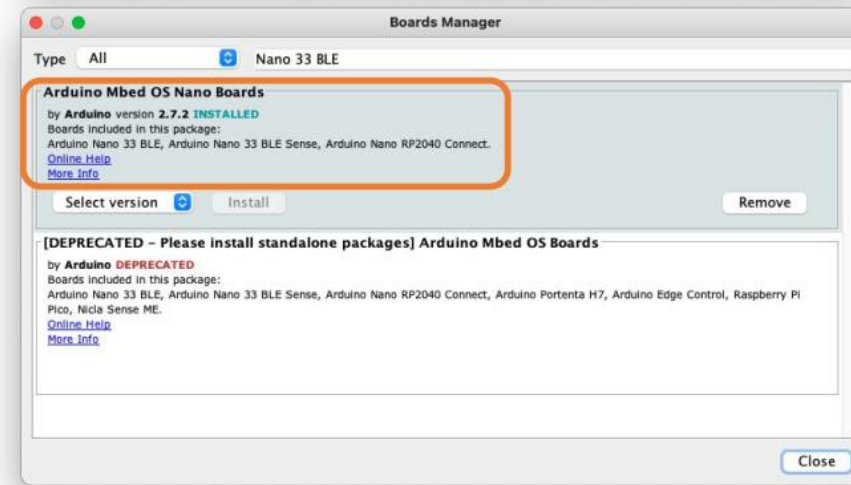
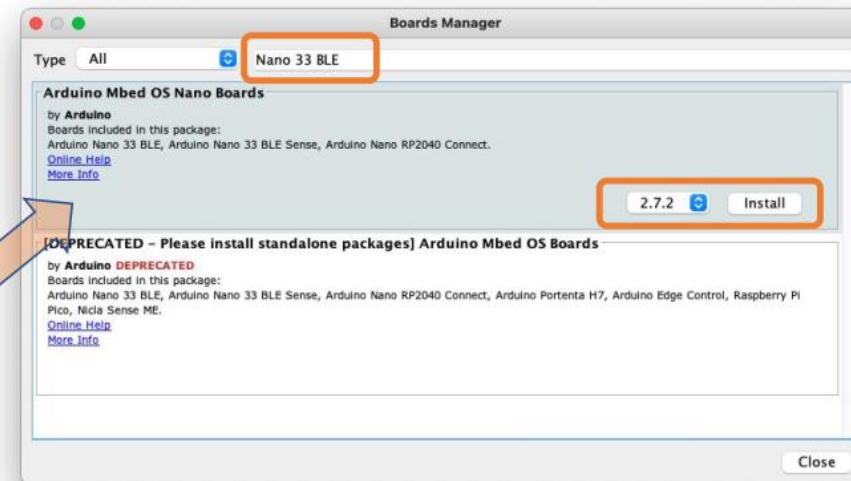
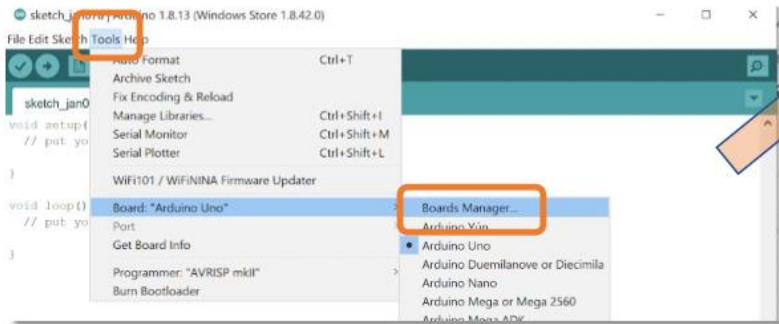


Área de
código

Console

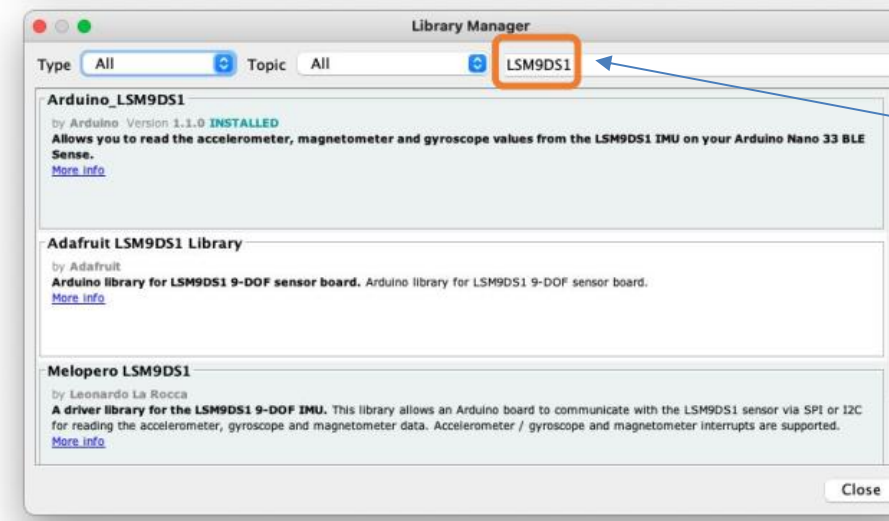
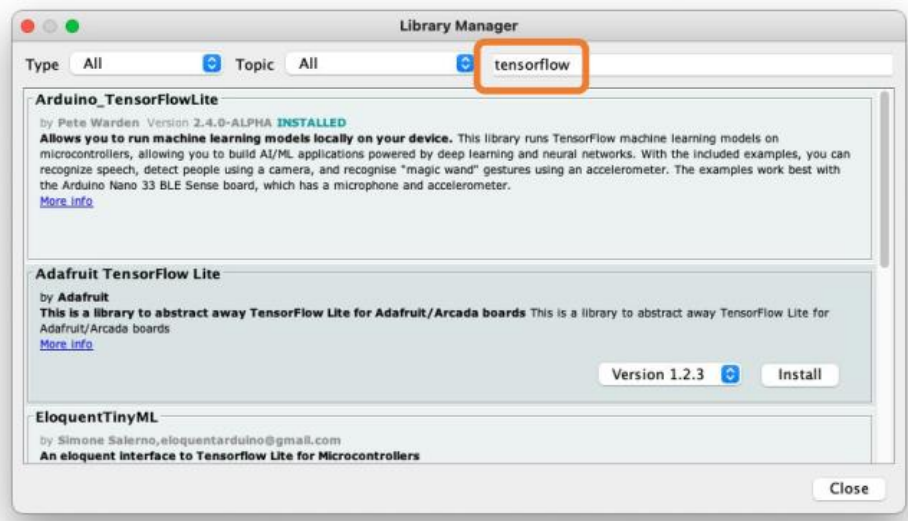
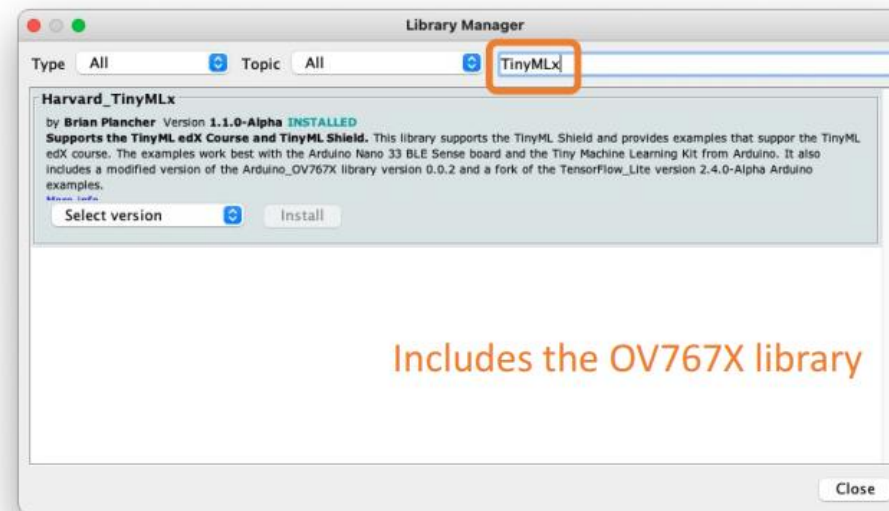
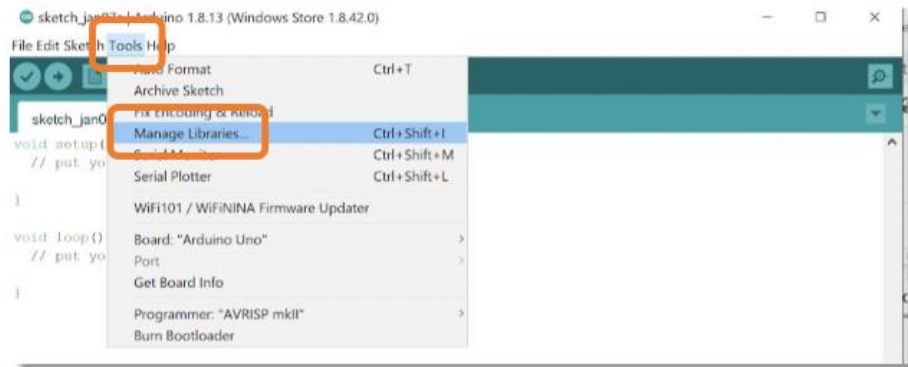
Arduino nano BLE 33 sense

Installing the Board Files



Arduino nano BLE 33 sense

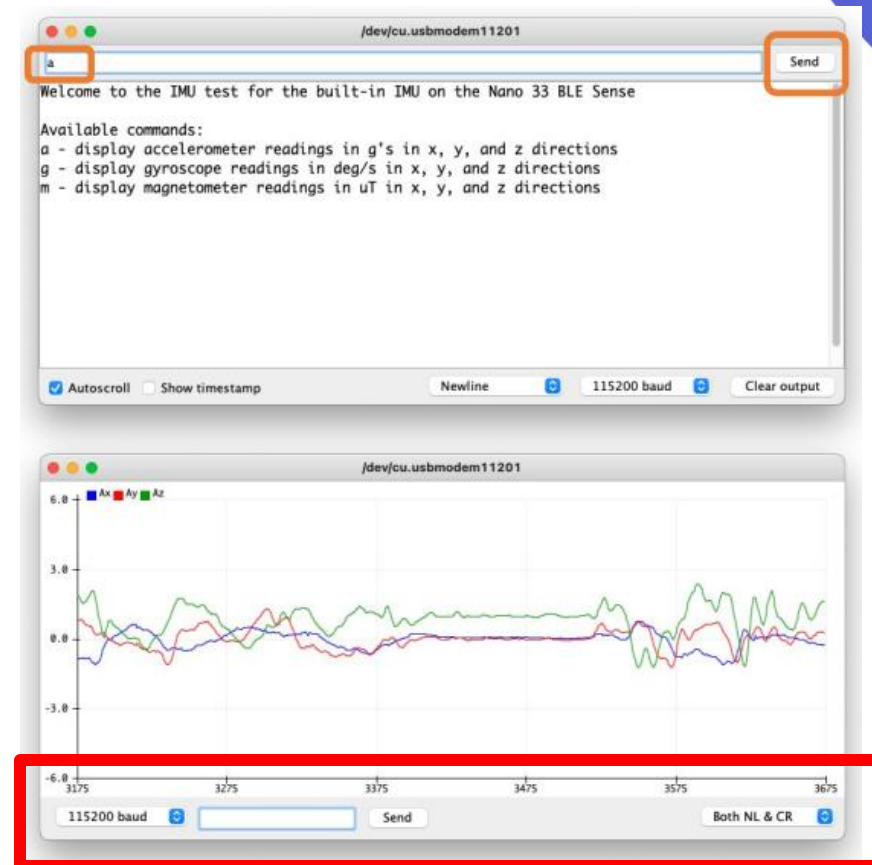
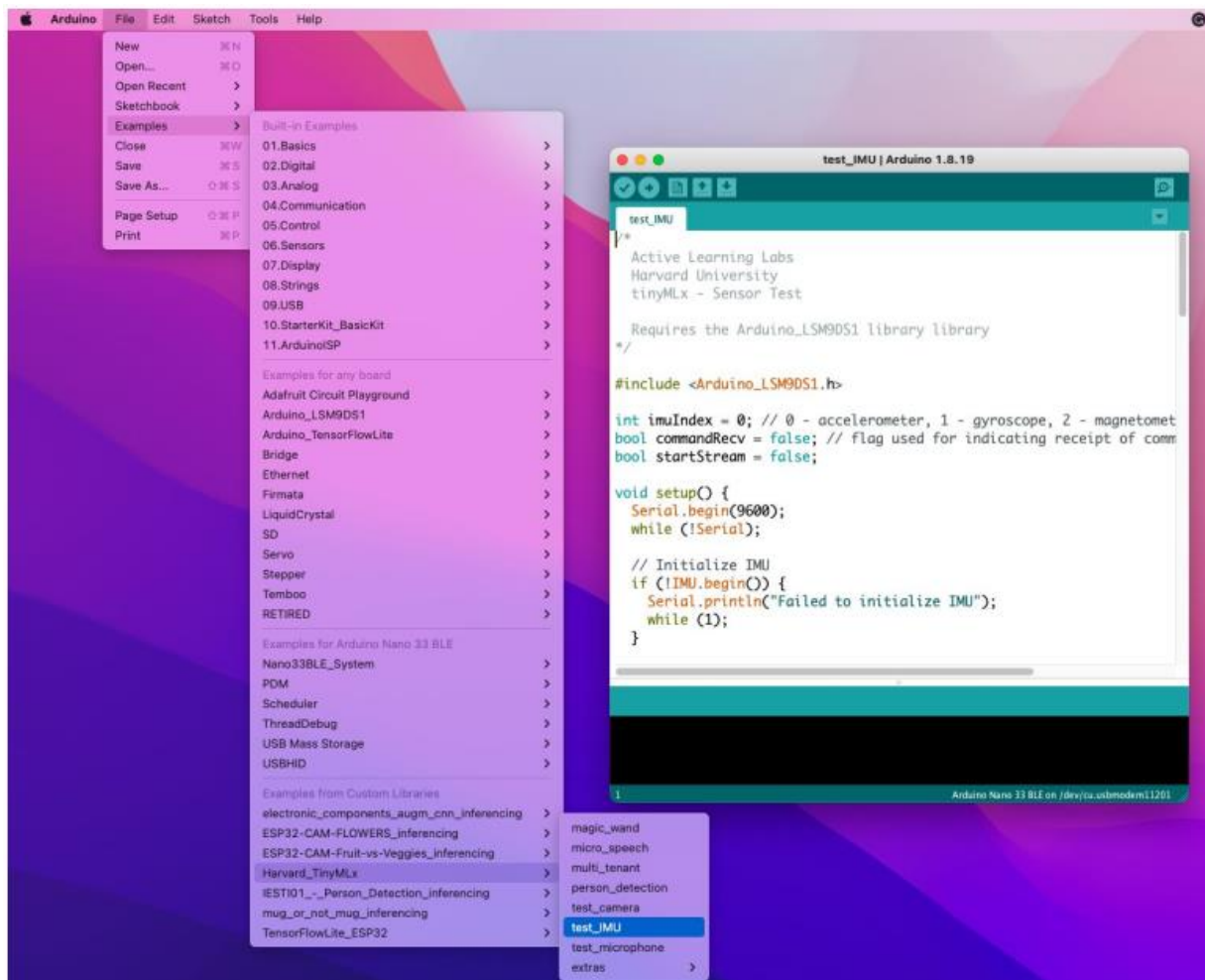
Installing the Main Libraries



Lib para
magnetômetro,
acelerômetro e
giroscópio.

Arduino nano BLE 33 sense

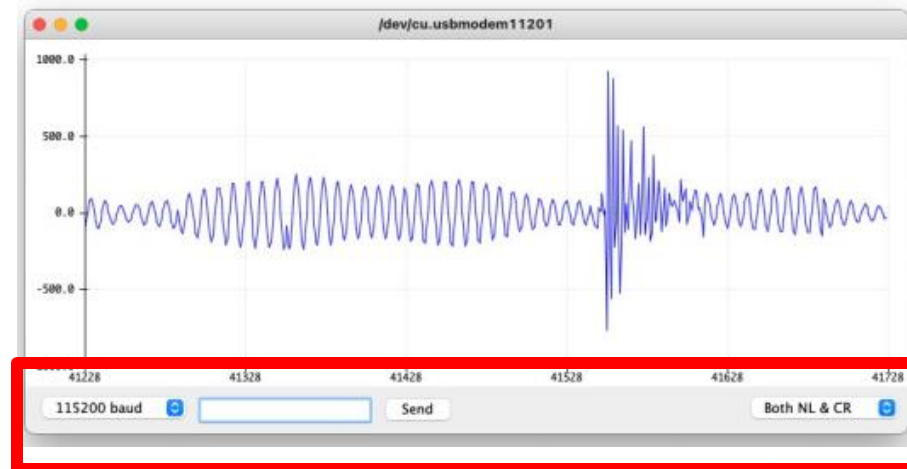
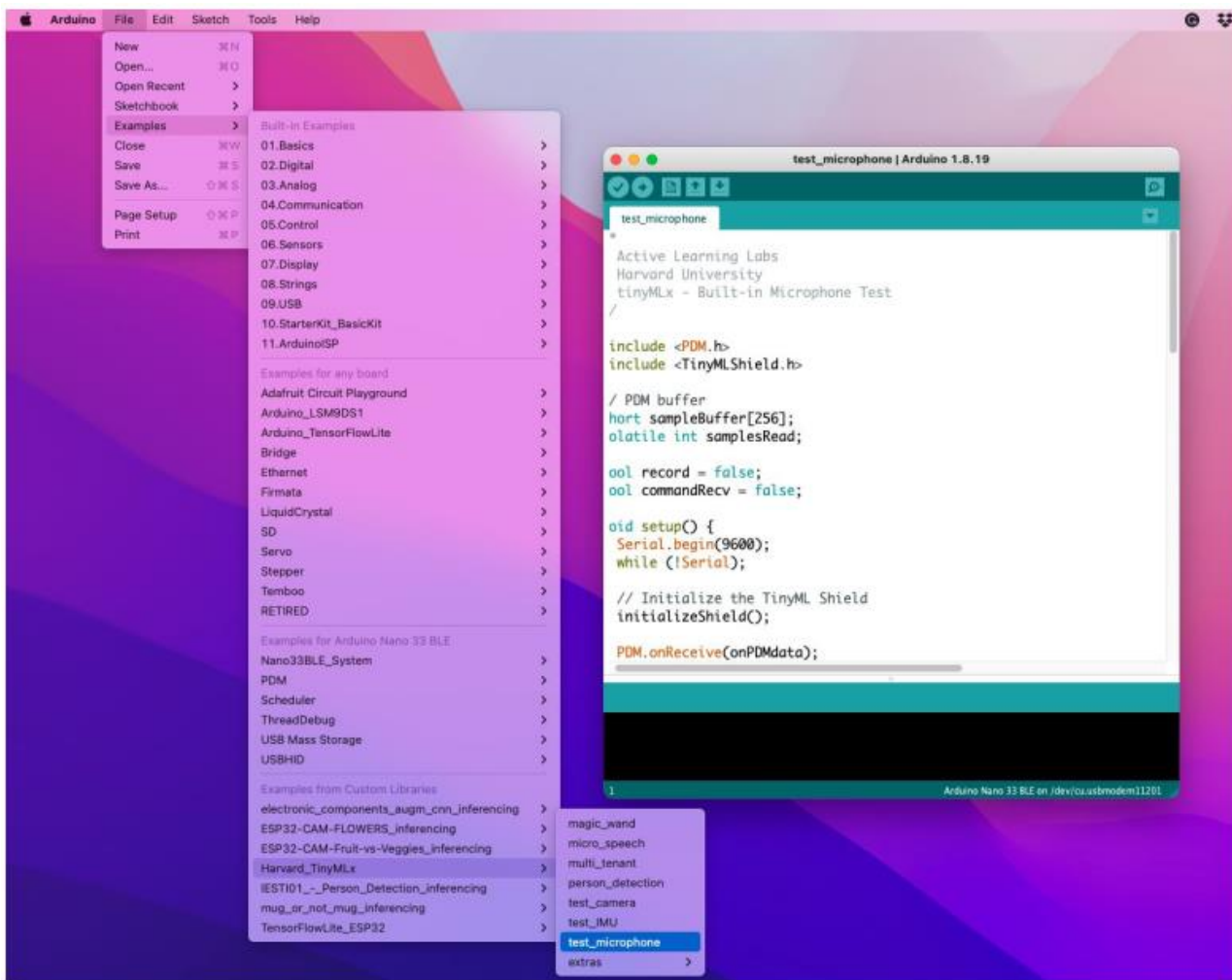
Exemplo: IMU



Notes: Close the Serial Monitor before open the Plotter
Repeat test for 'g' and 'm'

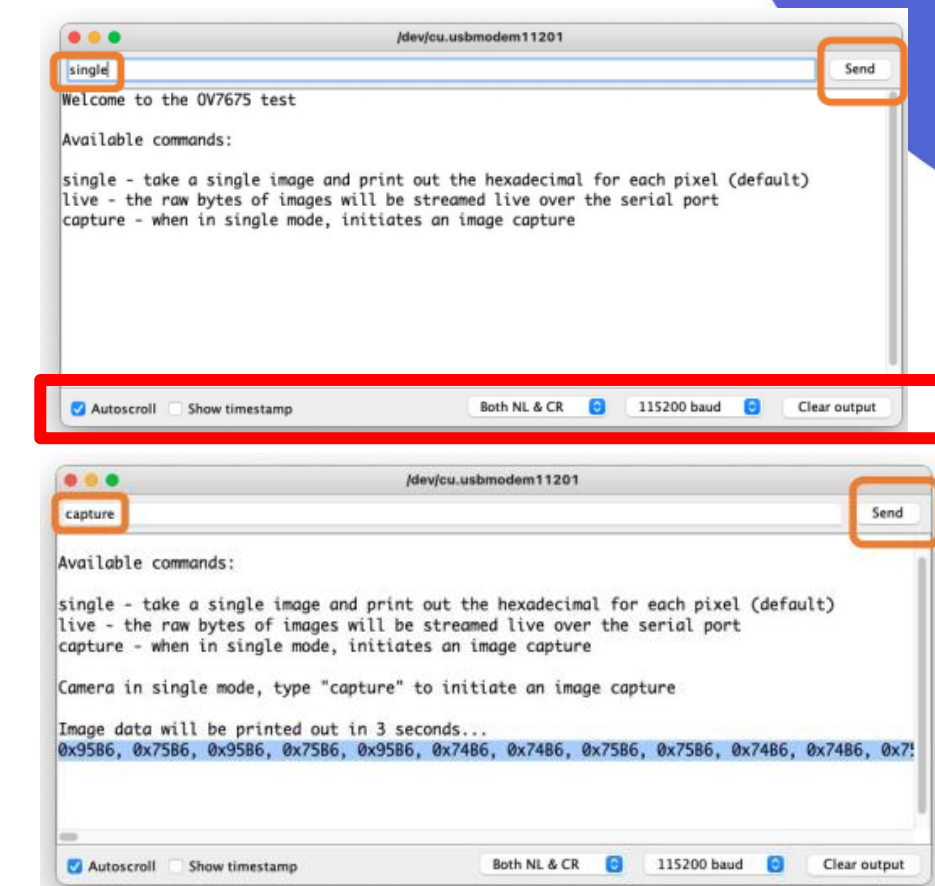
Arduino nano BLE 33 sense

Exemplo: Microfone



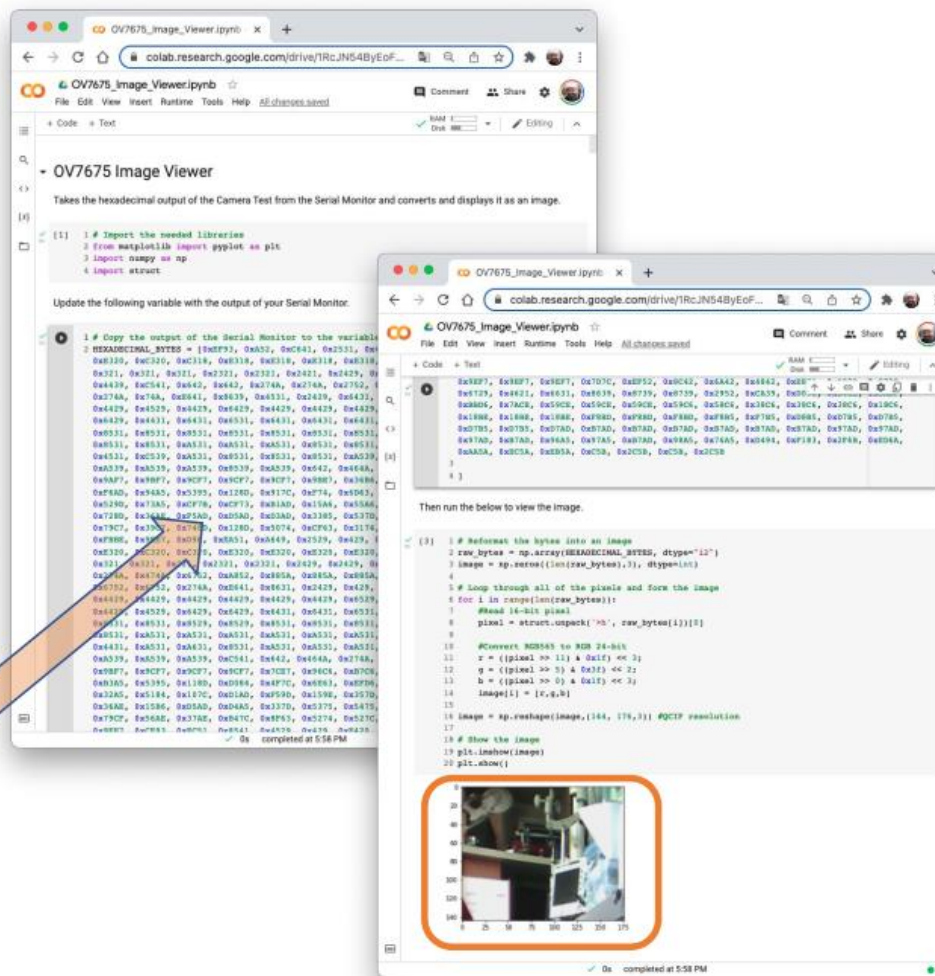
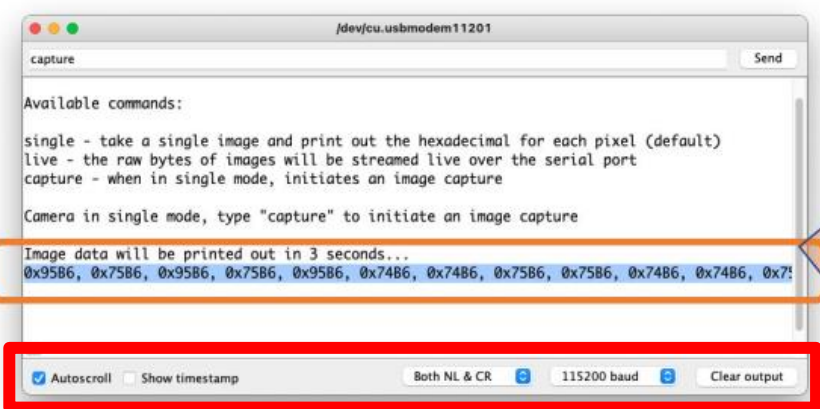
Note: Close the Serial Monitor before open the Plotter

Exemplo: Câmera



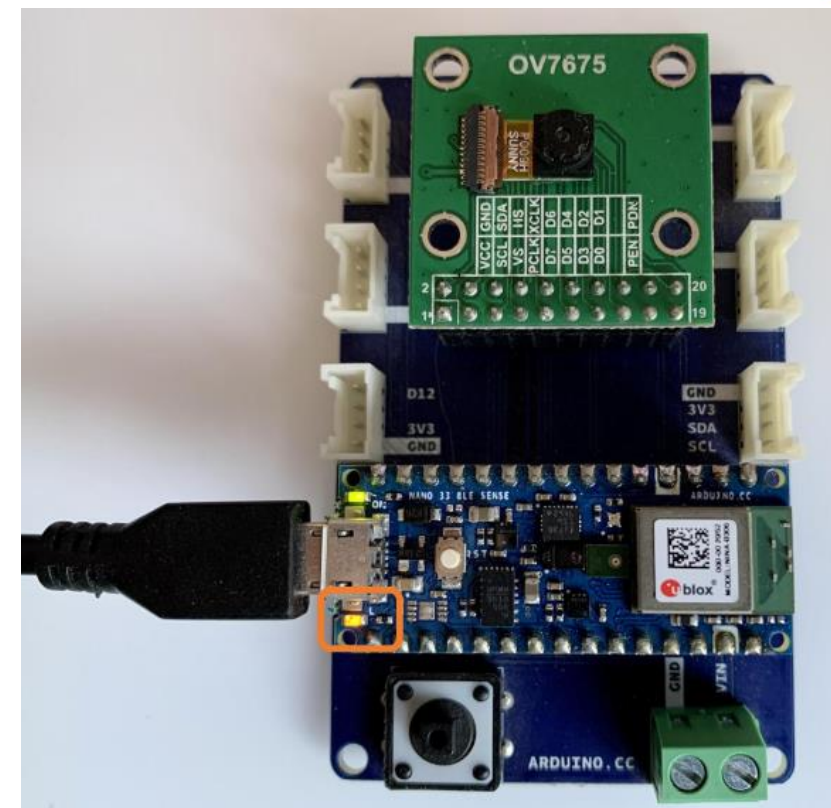
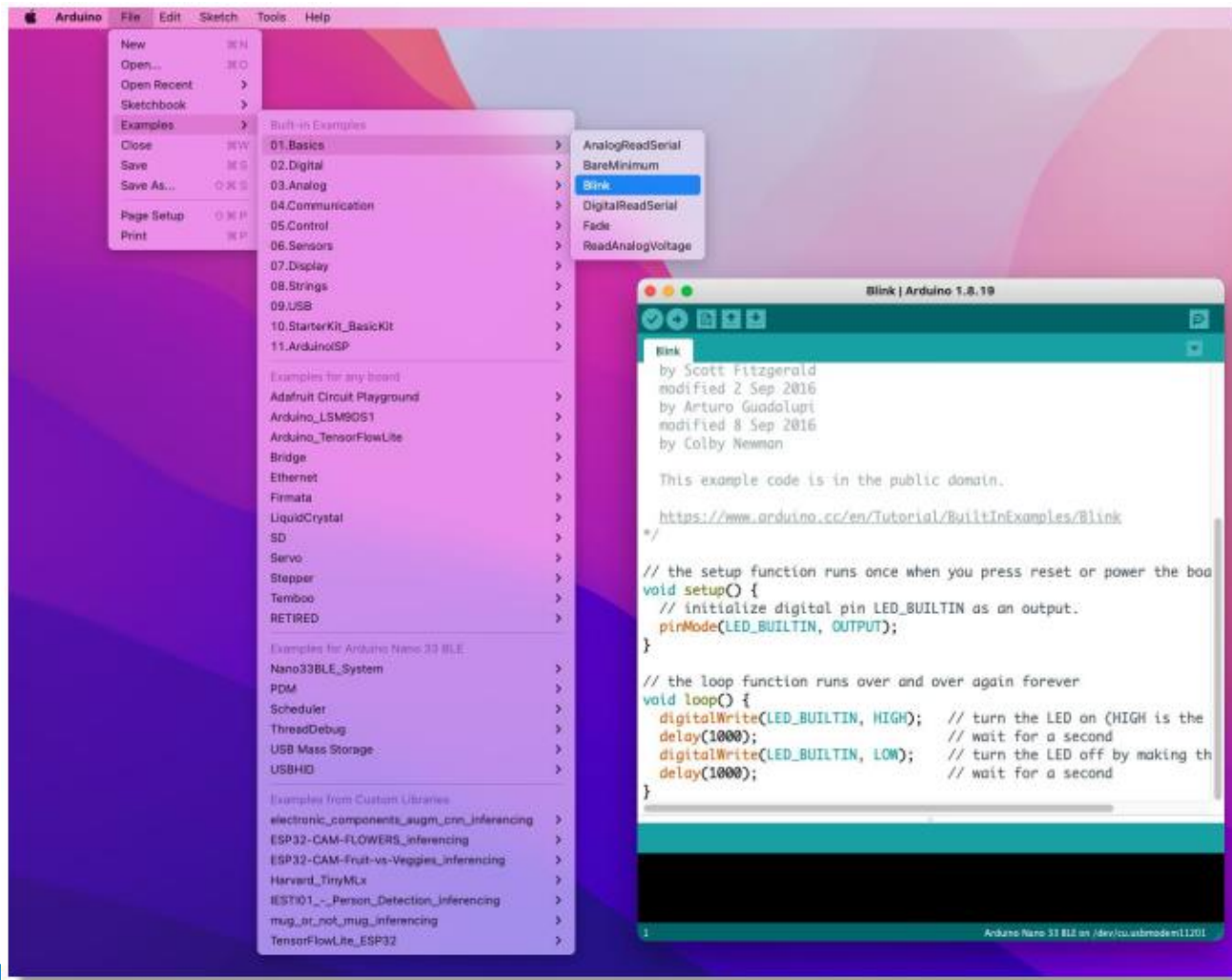
Note: You can Press Button insteady send 'capture'

Exemplo: Câmera



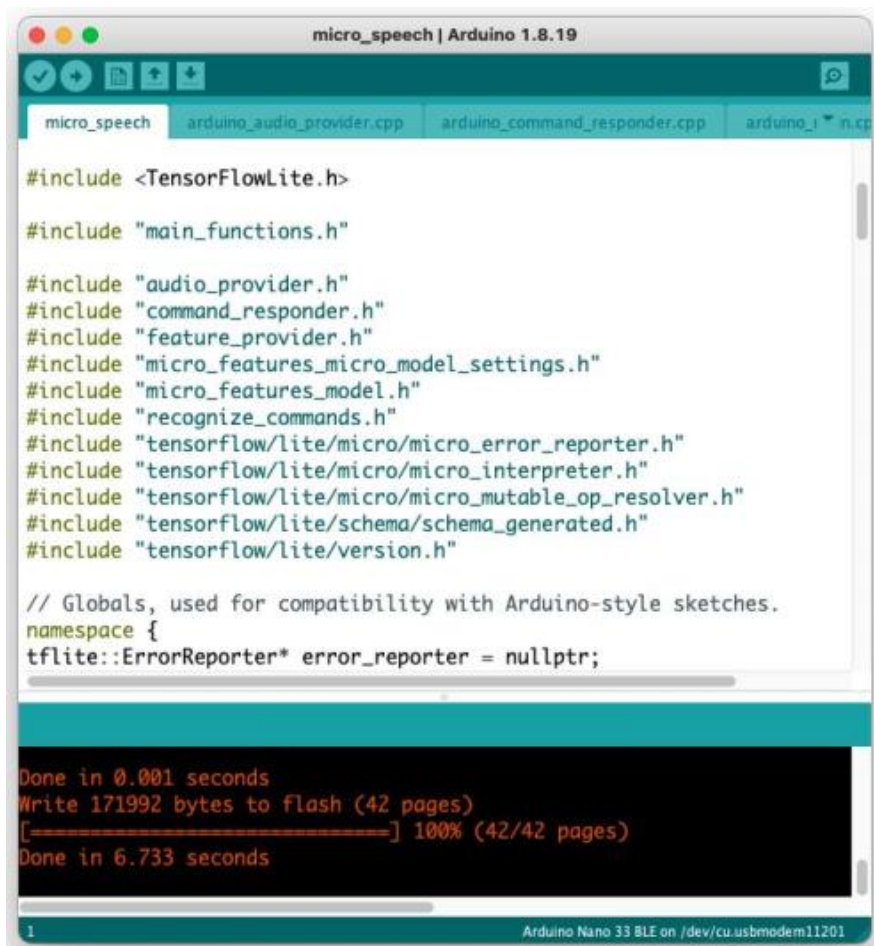
Arduino nano BLE 33 sense

Exemplo: Blink



Arduino nano BLE 33 sense

Exemplo: KeyWord Spotting (micro_speech)

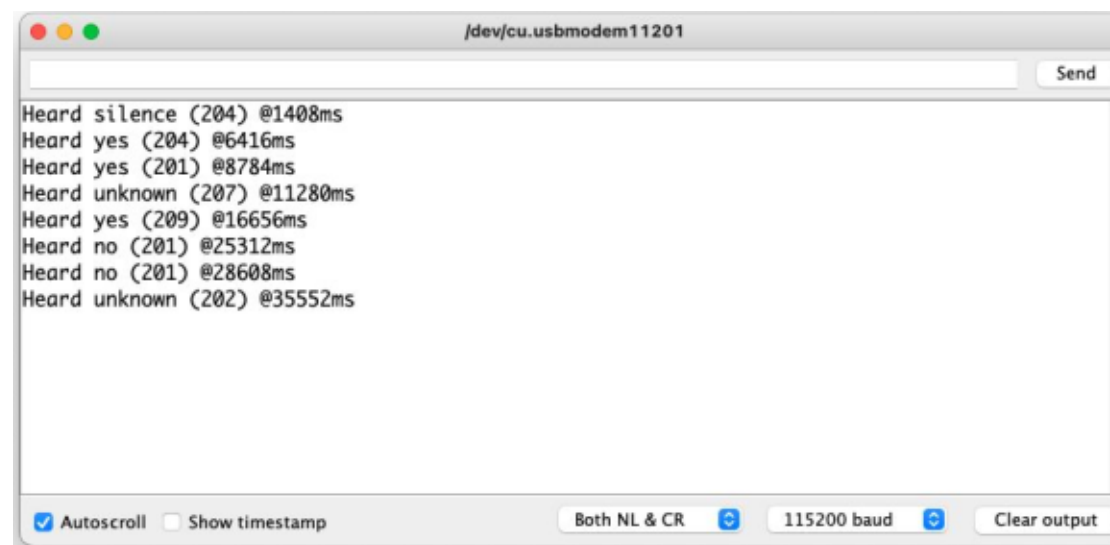
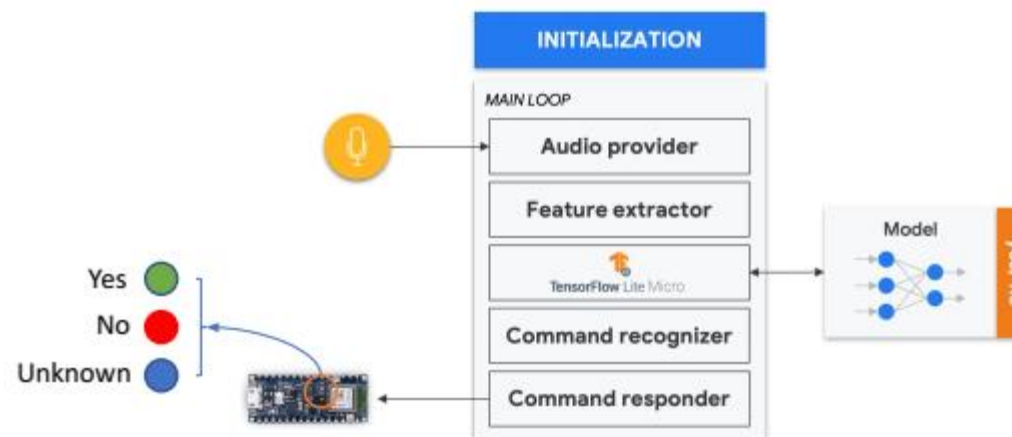


```
#include <TensorFlowLite.h>
#include "main_functions.h"

#include "audio_provider.h"
#include "command_responder.h"
#include "feature_provider.h"
#include "micro_features_micro_model_settings.h"
#include "micro_features_model.h"
#include "recognize_commands.h"
#include "tensorflow/lite/micro/micro_error_reporter.h"
#include "tensorflow/lite/micro/micro_interpreter.h"
#include "tensorflow/lite/micro/micro_mutable_op_resolver.h"
#include "tensorflow/lite/schema/schema_generated.h"
#include "tensorflow/lite/version.h"

// Globals, used for compatibility with Arduino-style sketches.
namespace {
  tflite::ErrorReporter* error_reporter = nullptr;

  Done in 0.001 seconds
  Write 171992 bytes to flash (42 pages)
  [-----] 100% (42/42 pages)
  Done in 6.733 seconds
```



```
/dev/cu.usbmodem11201
Send

Heard silence (204) @1408ms
Heard yes (204) @6416ms
Heard yes (201) @8784ms
Heard unknown (207) @11280ms
Heard yes (209) @16656ms
Heard no (201) @25312ms
Heard no (201) @28608ms
Heard unknown (202) @35552ms

Autoscroll Show timestamp Both NL & CR 115200 baud Clear output
```


Arduino nano BLE 33 sense

Exemplo: Person detection (person_detection)

```
person_detection | Arduino 1.8.19

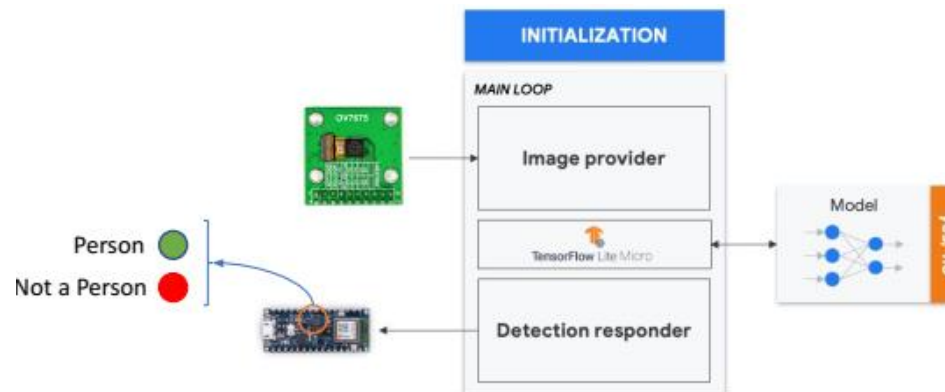
person_detection  arduino_detection_responder.cpp  arduino_image_provider.cpp  arduino_main.cpp

#include <TensorFlowLite.h>
#include "main_functions.h"
#include "detection_responder.h"
#include "image_provider.h"
#include "model_settings.h"
#include "person_detect_model_data.h"
#include "tensorflow/lite/micro/micro_error_reporter.h"
#include "tensorflow/lite/micro/micro_interpreter.h"
#include "tensorflow/lite/micro/micro_mutable_op_resolver.h"
#include "tensorflow/lite/schema/schema_generated.h"
#include "tensorflow/lite/version.h"

// Globals, used for compatibility with Arduino-style sketches.
namespace {
  tflite::ErrorReporter* error_reporter = nullptr;
  const tflite::Model* model = nullptr;
  tflite::MicroInterpreter* interpreter = nullptr;

  Done uploading.
Done in 0.001 seconds
Write 451984 bytes to flash (111 pages)
[=====] 100% (111/111 pages)
Done in 17.863 seconds

1 Arduino Nano 33 BLE on /dev/cu.usbmodem11201
```



```
/dev/cu.usbmodem11201

Person score: -37 No person score: 37
Person score: -39 No person score: 39
Person score: 10 No person score: -10
Person score: 2 No person score: -2
Person score: 0 No person score: 0
Person score: 22 No person score: -22
Person score: 22 No person score: -22
Person score: 21 No person score: -21
Person score: 9 No person score: -9
Person score: -1 No person score: 1
Person score: 14 No person score: -14
Person score: 12 No person score: -12
Person score: -40 No person score: 40
Person score: -34 No person score: 34

Autoscroll Show timestamp Both NL & CR 115200 baud Clear output
```


Edge Impulse

- Tarefa para casa
 - Execute um dos projetos criados na última aula no Arduino nano BLE 33 sense.